

New Business Section - Design Document

Phase 1: Deal Pipeline & Quick Evaluation

Date: August 12, 2026 **Status:** Design Review - Pending Team Input **Author:** Jim Bruin / Engineering

1. Overview

Purpose

The New Business section connects the front end of the deal sourcing and underwriting process to the existing asset management platform. Today, prospective deals are evaluated entirely in Excel — partner-provided Argus models, custom return spreadsheets, and manually assembled IC memos. When a deal closes, the underwritten data is re-keyed into MRI and the Projected IS file is manually loaded.

This design brings the evaluation workflow into the app so that:

- New deals can be quickly modeled using the same calculation engines that power asset management (waterfall, IRR, ROE, MOIC, debt service)
- Multiple scenarios can be saved and compared side-by-side
- When a deal closes, it onboards to the portfolio with one click — no re-keying
- The team has a shared pipeline view showing all prospective deals and their status
- Many more deals are evaluated than closed; the system accommodates high-volume screening

Scope (Phase 1)

Phase 1 focuses on two core capabilities:

1. **Deal Pipeline** — Track prospective deals through stages from Lead to Closed/Passed
2. **Quick Deal Evaluator** — "Does this pencil?" calculator using existing engines

Future phases will add rent roll analysis, lease testing, Excel cash flow import, IC memo generation, and cross-portfolio analysis.

2. Deal Pipeline

Pipeline Stages

Stage	Description
Lead	Opportunity identified; minimal info (name, location, price, broker)
Screening	Quick evaluation run; initial returns assessment
LOI	Letter of intent submitted or under negotiation
Due Diligence	PCA, environmental, title, legal, empire analysis underway

IC Review	Investment committee memo prepared and under review
Closing	Approved; legal documentation and funding in progress
Closed	Transaction completed; ready to onboard to portfolio
Passed	Deal declined (with reason captured)

Deal Record Fields

Field	Type	Required	Notes
Deal Name	Text	Yes	e.g., "Vestavia Hills City Center"
Location	Text	Yes	City, State
Asset Type	Select	Yes	Retail, Multifamily, Office, Industrial, Mixed-Use
GLA (SF)	Number	No	Gross leasable area
Units	Number	No	For multifamily
Year Built	Number	No	
Partner Name	Text	No	e.g., "Burton Property Group"
Stage	Select	Yes	See stages above
Assigned To	Select	No	Team member
Target Close Date	Date	No	
Purchase Price	Currency	No	Can be entered later
Source / Broker	Text	No	

Pass Reason	Text	No	Required when stage = Passed
Notes	Text	No	Free-form

Pipeline Views

Kanban Board — Cards organized by stage in columns. Drag-and-drop to advance stage. Cards show deal name, location, purchase price, and partner. Color-coded by asset type.

Table View — Sortable, filterable list with all fields. Toggle between Kanban and Table with a button. Default sort by last updated.

Activity Log — Every deal has a chronological log of stage changes, notes, and evaluation runs. Visible in the deal workspace.

3. Quick Deal Evaluator

Concept

A single-page workspace where the user enters deal assumptions on the left and sees instant computed returns on the right. This replaces the Summary + Returns sheets from the current Excel workflow.

The evaluator uses the app's existing calculation engines — the same code that powers Deal Analysis, waterfall distributions, and partner returns. No separate calculation logic to maintain.

Input Panel (Left Side)

Acquisition

Field	Type	Default	Notes
Purchase Price	Currency	—	Required
Closing Costs	% or \$	2%	Toggle between percentage and fixed dollar
CapEx at Close	Currency	0	Upfront capital expenditure budget
Total Cost Basis	Computed	—	Purchase + Closing Costs + CapEx (auto-calculated)

Debt Assumptions

Field	Type	Default	Notes
Loan Amount	\$ or LTV%	65%	Toggle between dollar amount and LTV percentage

Interest Rate	%	—	Fixed rate, OR enter as UST Yield + Spread
Term	Years	7	Loan term
I/O Period	Months	60	Interest-only period
Amortization	Years	30	After I/O period
Origination Fee	%	0.25%	Of loan amount

Equity Structure

Field	Type	Default	Notes
PSC Equity %	%	90%	PSC share of total equity
Partner Equity %	%	10%	Auto-calculated as complement
Partner Name	Text	—	Pre-filled from deal record

Partnership Terms

Field	Type	Default	Notes
Preferred Return	%	8.0%	Cumulative compounded annual pref rate
PSC Promote	%	20%	PSC participation after pref is current
AM Fee	%	0.95%	Annual fee on invested PSC capital
Annual Expenses	\$	7,500	Venture/partnership expenses
CF Distribution	Select	Pari-passu	How pref is distributed (pari-passu or PSC priority)

Operating Assumptions

Field	T	C	Notes
Year 1 NOI			Current Required
NOI Growth Rate			Annual growth % 2.0% Applied to NOI
CapEx Reserve			Annual \$500k Replacement reserves
— OR —			

```
Table Override:
enter
NOI
per
year
manually
```

Exit Assumptions

Results Panel (Right Side)

Key Metrics (KPI Cards)

Annual Summary Table

Year	Net Sales	Cost of Sales	Gross Profit	Operating Expenses	Operating Profit	Pre-tax Income	Income Tax	Net Income	ROE
1	...	...	...	...	...	...	...	...	...
2	...	...	...	...	...	...	...	...	...
...									
Exit								Sale Proceeds	

Capital Stack Visualization

Horizontal stacked bar showing: - Debt (blue) with LTV % - PSC Preferred Equity (green) with exposure % - Partner Equity (gray) - PSC Exposure line (Debt + PSC PE as % of value)

Sensitivity Table

Matrix showing PSC IRR at different combinations of: - **Rows:** Exit Cap Rate (e.g., 6.5%, 7.0%, 7.5%, 8.0%, 8.5%) -

Columns: Hold Period (e.g., 5yr, 6yr, 7yr, 8yr, 9yr, 10yr)

Highlights the base case cell. Color-codes cells (green > target IRR, yellow near target, red below).

Cash Flow Chart

Dual-axis chart: - Bars: Annual NOI - Line: PSC cash flow (cumulative or annual) - Marker: Capital event (sale proceeds)

Scenarios

Users can save multiple assumption sets per deal:

- **"Base Case"** — primary underwriting
- **"Downside - AMC Vacates"** — stress test
- **"7-Year Hold"** vs **"10-Year Hold"** — hold period comparison
- **"Standalone"** vs **"Crossed Portfolio"** — future phase

Each scenario saves all inputs and computed results. A comparison view shows scenarios side-by-side with delta highlighting.

4. Engine Integration

How Existing Code Is Reused

The Quick Evaluator does NOT duplicate calculation logic. It builds the same data structures that `compute_deal_analysis()` expects, populated from form inputs instead of database tables:

App Engine	Evaluator Usage
<code>Loan model (models.py)</code>	Built from debt assumptions (rate, term, I/O, amort)
<code>build_loan_schedule()</code> (loans.py)	Generates amortization schedule for debt service
<code>InvestorState (models.py)</code>	Tracks PSC and partner capital, pref accrual
Waterfall steps (waterfall.py)	Dynamically built from partnership terms (Pref, Share, Promote)
<code>run_interleaved_waterfalls()</code> (compute.py)	Runs CF + Capital waterfalls on synthetic cash flows
<code>build_partner_results()</code> (compute.py)	Computes IRR, ROE, MOIC per partner
<code>xirr()</code> (metrics.py)	IRR calculation with irregular dates
<code>calculate_roe_detailed()</code> (metrics.py)	Weighted average capital ROE
<code>build_cash_flow_schedule_from_fad()</code>	Cash management / distributable cash

projected_cap_rate_at_date()	Exit valuation
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A new function, `build_prospect_analysis()`, will:

1. Generate a synthetic forecast DataFrame from NOI inputs (Year 1 + growth, or manual schedule)
2. Create a `Loan` object from debt assumptions
3. Build waterfall steps from partnership terms
4. Call `compute_deal_analysis()` with these synthetic inputs
5. Return the same result structure that Deal Analysis uses

This means the Evaluator's returns will always match what Deal Analysis shows after the deal is onboarded.

5. Onboarding to Portfolio (Deal Closes)

When a prospect's stage is set to "Closed," a wizard converts it to a portfolio deal:

Step	Action	Target Table
1	Assign vcode (auto or manual)	—
2	Create deal record	inv (deals)
3	Create waterfall steps	waterfalls
4	Create loan record	loans
5	Generate Projected IS entries	forecast_feed or isbs_projected_is
6	Create entity relationships	relationships
7	Link prospect to portfolio deal	prospect_deals.onboarded_vcode

The wizard pre-fills all fields from the prospect's assumptions. The user reviews and adjusts before confirming. After onboarding, the deal appears in Dashboard, Deal Analysis, One Pager, and all reports.

6. Data Architecture

New Database Tables

prospect_deals	
id	INTEGER PRIMARY KEY AUTOINCREMENT
deal_name	TEXT NOT NULL
location	TEXT
asset_type	TEXT
gla_sf	REAL
units	INTEGER
year_built	INTEGER
partner_name	TEXT
source_broker	TEXT
assigned_to	TEXT
stage	TEXT DEFAULT 'lead'
pass_reason	TEXT
target_close	TEXT (ISO date)

purchase_price	REAL
closing_cost_pct	REAL DEFAULT 0.02
capex_at_close	REAL DEFAULT 0
notes	TEXT
onboarded_vcode	TEXT (links to inv after close)
created_by	TEXT
created_at	TEXT
updated_at	TEXT

prospect_assumptions	
id	INTEGER PRIMARY KEY AUTOINCREMENT
prospect_id	INTEGER REFERENCES prospect_deals(id)
version	INTEGER DEFAULT 1
version_label	TEXT DEFAULT 'Base Case'
debt_amount	REAL
debt_rate	REAL
debt_term_months	INTEGER DEFAULT 84
io_months	INTEGER DEFAULT 60
amort_months	INTEGER DEFAULT 360
origination_fee	REAL DEFAULT 0.0025
psc_equity_pct	REAL DEFAULT 0.90
pref_rate	REAL DEFAULT 0.08
promote_pct	REAL DEFAULT 0.20
am_fee_pct	REAL DEFAULT 0.0095
annual_expenses	REAL DEFAULT 7500
exit_cap_rate	REAL
selling_cost_pct	REAL DEFAULT 0.02
hold_years	INTEGER DEFAULT 7
capex_reserve_psf	REAL DEFAULT 0.80
noi_year1	REAL
noi_growth_rate	REAL DEFAULT 0.02
crossed_vcodes	TEXT (future: comma-separated vcodes)
created_at	TEXT
updated_at	TEXT

prospect_cashflows	
id	INTEGER PRIMARY KEY AUTOINCREMENT
prospect_id	INTEGER REFERENCES prospect_deals(id)
version	INTEGER DEFAULT 1
period_date	TEXT (ISO date)
revenue	REAL
expenses	REAL
noi	REAL
capex	REAL
other	REAL
source	TEXT DEFAULT 'manual'

prospect_activity	
id	INTEGER PRIMARY KEY AUTOINCREMENT
prospect_id	INTEGER REFERENCES prospect_deals(id)
username	TEXT
action	TEXT (created, stage_change, note, evaluated, etc.)
note	TEXT
created_at	TEXT

All prospect tables will be added to PROTECTED_TABLES to prevent accidental overwrite during CSV imports.

API Endpoints

Pipeline:

GET	/api/prospects	List all (filterable by stage, assigned_to)
POST	/api/prospects	Create new prospect
GET	/api/prospects/<id>	Get detail
PUT	/api/prospects/<id>	Update (including stage changes)
DELETE	/api/prospects/<id>	Archive/delete

Assumptions & Evaluation:

GET	/api/prospects/<id>/assumptions	List scenarios
POST	/api/prospects/<id>/assumptions	Save new scenario
PUT	/api/prospects/<id>/assumptions/<v>	Update scenario
POST	/api/prospects/<id>/evaluate	Run returns calculation
POST	/api/prospects/<id>/evaluate/excel	Download results as Excel
Cash Flow Import (Future Phase):		
POST	/api/prospects/<id>/import-cf	Upload Excel cash flow
Onboarding:		
POST	/api/prospects/<id>/onboard	Convert to portfolio deal
Activity:		
GET	/api/prospects/<id>/activity	Activity log
POST	/api/prospects/<id>/activity	Add note

7. User Interface Layout

Pipeline View (default landing for New Business)

+-----+ New Business [+ New Deal]					
+-----+ [Kanban] [Table] [Filter: Stage] [Assigned To]					
+-----+ Lead Screening LOI DD IC ...					
+-----+ +-----+ +-----+					
+-----+ Vestavia Park Pl Meridian					
+-----+ Birm,AL Atl, GA Hou, TX					
+-----+ \$76M \$42M \$31M					
+-----+ Burton CBRE Greystar					
+-----+ +-----+ +-----+					
+-----+					
+-----+					

Deal Workspace (click into a deal)

+-----+ < Back to Pipeline Vestavia Hills City Center [Passed v]					
+-----+ Birmingham, AL Retail 389,471 SF Burton Property Group					
+-----+ ASSUMPTIONS RESULTS					
+-----+ Acquisition PSC IRR PSC ROE PSC MOIC					
+-----+ Purchase Price \$76,000,000 [13.2%] [9.3%] [2.02x]					
+-----+ Closing Costs 2.0%					
+-----+ CapEx at Close \$4,000,000 Inv IRR Inv ROE Inv MOIC					
+-----+ Total Basis \$82,700,000 [11.3%] [8.2%] [1.84x]					
+-----+ Debt Property IRR (Unlev) [9.4%]					
+-----+ LTV 65% Property IRR (Lev) [13.9%]					
+-----+ Rate 6.15% Ops/Residual 62% / 38%					
+-----+ Term 7 yrs					
+-----+ I/O 60 mo Annual Summary					
+-----+ Amort 30 yr +-----+					
+-----+ Yr NOI CFAD ROE					
+-----+ Equity 1 5.6M 5.2M 10.4%					
+-----+ PSC 90% 2 6.1M 5.4M 8.5%					
+-----+ Partner 10% 3 6.4M 5.5M 8.5%					
+-----+ 					
+-----+ Partnership Terms +-----+					

8. What This Replaces

9. Future Phases

- Upload partner-provided Excel/Argus models
- Auto-detect and map revenue, expense, NOI rows
- Monthly cash flow support (not just annual)

- Tenant-level revenue assumptions
- CapEx schedule (itemized, not just \$/SF)

Phase 3: Rent Roll & Lease Testing

- Import rent roll from Excel
- Lease expiration analysis and rollover schedule
- Mark-to-market analysis (in-place vs market rents)
- Vacancy and downtime modeling
- Recovery/reimbursement audit

Phase 4: IC Memo Generation

- Auto-populate memo template from deal data
- Sources & Uses table from assumptions
- Returns summary with sensitivity
- Capital stack diagram
- Export to formatted PDF

Phase 5: Cross-Portfolio Analysis

- Cross a prospect with existing portfolio deals
- Combined capital accounts and waterfalls
- Portfolio-level exposure analysis
- Matches the "Four Pack" scenario from the current Excel

Phase 6: Term Sheet Generator

- Generate term sheets from partnership terms
- Standard templates by deal type
- Version tracking and approval workflow

10. Implementation Priority

Priority	Component	Effort	Dependencies
1	Database tables + API (CRUD)	1 sprint	None
2	Pipeline view (Kanban + Table)	1 sprint	API
3	Quick Evaluator (form + engine)	2 sprints	API + existing engines
4	Scenario management	0.5 sprint	Evaluator

5	Excel export of results	0.5 sprint	Evaluator
6	Onboard wizard	1 sprint	Evaluator + all tables

11. Questions for the Team

Pipeline stages — Are the 8 stages (Lead through Closed/Passed) right? Are any missing? Should there be sub-stages (e.g., DD has environmental, title, legal)?

Default assumptions — Are these defaults reasonable for most deals? PSC 90%, Pref 8%, Promote 20%, AM Fee 0.95%, Expenses \$7,500, Closing 2%, Selling 2%, LTV 65%, Hold 7yr, Growth 2%, CapEx \$0.80/SF.

Evaluation frequency — Do you re-run returns frequently as you refine assumptions, or is it more of a "run once, save, move on" workflow? This affects whether we auto-compute on every field change or use an explicit "Evaluate" button.

Scenario naming — What are the most common scenarios you model? (e.g., Base/Downside/Upside, different hold periods, crossed vs standalone, different exit caps)

Who accesses this? — Should all users see the pipeline, or only New Business team members? Should analysts be able to edit assumptions, or view-only?

Crossing timing — At what stage do you typically model the crossed portfolio returns? Is it always at IC stage, or earlier during screening?

What data from the Excel model do you wish you didn't have to re-enter? What are the biggest pain points in the current process?